Room Environment V3

Founder Summary Report

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Executive Summary

The current room environment pipeline is not failing because Gemini is weak. It is failing because the code and process around Gemini are too loose.

Today the system simplifies rooms too aggressively, treats biomes mostly like flavor text, and asks the model for too much in one step. That creates art that can look interesting in isolation but still does not read correctly as a wall, floor, platform, door, or room shell.

The proposed v3 pipeline keeps Gemini and the schema-plus-template approach, but changes the rest of the pipeline so the model is working inside tighter production rules.

What Changes In Plain English

- The room layout becomes the authority. Planning must fit real geometry, doors, major platforms, pits, ceiling shape, and traversal routes.
- Biomes become real production data. Each biome gets explicit visual rules, identity checks, and canonical style anchors.
- Structural and scenic art are separated. Walls, floors, ceilings, platforms, and thresholds must read clearly on their own.
- Gemini works slot by slot. The system stops asking for one giant all-in-one room answer.
- Human review becomes a hard gate. QA and Creative review screenshots from the workflow and runtime over several rounds.

Main Problems Found

- Rooms are reduced too aggressively before generation, so output does not match real traversal structure.
- Biomes are not meaningfully selected per room, so results drift toward one shared look.
- Prompting is too monolithic, and fallback logic hides model weaknesses instead of fixing them.
- The workflow does not strongly protect component-fit, which is why some outputs look nice but fail their gameplay role.
- Runtime review is too weak on its own and must be backed by manual screenshot review.

Stakeholder Review Outcome

- Engineering supports v3, but wants clear schema versioning and module boundaries before coding.
- QA supports v3, but wants explicit blocker criteria, fixed fixtures, and runtime-first evidence.
- Creative supports v3, but wants stronger biome identity rules and rejection criteria for component-fit failures.
- Design supports v3, but wants proposal-first theming and a fixed review-surface order.
- Game Director supports v3, but wants room role and progression context added to planning.

Key Changes To Approve

- Explicit biome definitions with identity checklists, forbidden motifs, and canonical style anchors.
- A split environment spec: room intent, component contracts, assembly plan, and review state.
- Proposal-first biome, palette, and environment application until a user accepts them.
- Mandatory assembly-plan visibility, runtime-first review, QA blocker criteria, and Creative rejection codes.
- A narrow first slice: one biome, three fixture rooms, and repeated QA plus Creative review rounds.